

HIKONE, Japan

WMO: 477610

Lat: 35.2758N

Lon: 136.2436E

Elev: 304

StdP: 14.53

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.7	31.2	19.7	15.0	37.0	22.0	16.7	37.2	24.5	36.6	22.4	38.1	7.5	150	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	92.1	78.0	90.2	77.5	88.2	76.7	79.6	88.9	78.6	87.5	77.6	85.9	7.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	141.8	85.3	76.0	137.2	84.1	75.1	133.2	83.1	43.4	89.1	42.3	87.7	41.4	86.2	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.2	16.0	DB	26.2	95.6	2.1	1.7	24.6	96.8	23.4	97.9	22.2	98.8	20.7	100.1	(4)
(5)			WB	24.4	81.5	2.0	1.0	23.0	82.2	21.8	82.7	20.7	83.3	19.2	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	38.9	39.7	45.2	54.9	64.4	71.8	79.9	82.5	75.0	64.3	53.1	43.7	(6)	
	DBStd	15.74	3.66	4.28	5.09	5.69	4.52	4.13	4.23	3.41	4.93	4.94	5.04	4.71	(7)	
	HDD50	1049	345	291	165	19	0	0	0	0	0	0	25	204	(8)	
	HDD65	3600	810	710	613	306	66	2	0	0	1	72	358	662	(9)	
	CDD50	4536	0	1	17	165	447	653	927	1007	751	444	117	7	(10)	
	CDD65	1611	0	0	0	2	48	205	462	542	302	50	0	0	(11)	
	CDH74	13421	0	0	0	11	174	903	4285	5921	2011	116	0	0	(12)	
	CDH80	4501	0	0	0	1	14	152	1496	2340	490	8	0	0	(13)	
(14) Wind	WSAvg	6.3	7.6	7.6	7.1	6.0	5.5	5.0	5.1	5.5	6.0	6.2	6.4	7.2	(14)	
(15) Precipitation	PrecAvg	71.7	6.4	4.9	4.9	4.4	5.9	7.0	8.7	5.5	7.5	5.3	4.8	6.5	(15)	
	PrecMax	89.4	10.6	6.9	7.3	7.8	16.0	12.7	16.2	16.8	15.9	10.9	8.5	12.5	(16)	
	PrecMin	51.0	2.4	1.8	2.4	1.5	2.4	3.0	2.3	1.2	2.6	1.7	2.2	3.3	(17)	
	PrecStd	8.6	2.2	1.3	1.4	1.7	3.2	2.4	4.0	3.5	3.5	2.1	1.7	2.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.6	56.0	65.4	75.2	81.9	87.7	93.9	94.6	90.4	80.5	68.2	59.6	(19)	
		MCWB	46.3	49.6	54.4	60.5	65.6	74.2	78.5	78.5	76.3	71.0	60.4	54.4	(20)	
	2%	DB	49.1	51.7	60.3	70.7	78.4	84.0	91.6	92.7	87.5	77.0	65.4	56.0	(21)	
		MCWB	44.0	46.1	51.5	58.1	63.8	72.0	77.8	78.3	75.7	68.2	58.8	51.1	(22)	
	5%	DB	47.1	49.1	56.8	67.4	75.5	81.2	89.6	91.1	85.1	74.6	63.5	53.6	(23)	
		MCWB	42.5	44.5	50.0	57.3	63.0	70.6	77.3	77.9	74.5	66.2	57.7	48.8	(24)	
	10%	DB	45.4	47.1	53.6	64.4	72.8	79.0	87.3	89.4	82.7	72.3	61.6	51.5	(25)	
		MCWB	41.1	42.8	47.9	56.0	62.1	69.9	76.6	77.4	73.4	64.4	56.3	46.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.1	51.4	56.6	65.0	70.0	76.8	80.8	81.1	78.9	72.8	63.4	56.1	(27)	
		MCDB	50.1	54.2	61.6	69.1	76.2	84.7	90.5	90.9	87.1	77.9	65.8	58.3	(28)	
	2%	WB	45.2	47.6	53.8	61.7	67.2	74.9	79.1	80.0	77.1	70.0	61.0	52.2	(29)	
		MCDB	47.8	50.2	58.6	67.8	73.7	80.8	88.3	89.5	84.4	75.2	63.8	55.0	(30)	
	5%	WB	43.3	45.5	51.2	59.0	65.2	73.3	78.2	79.1	75.9	67.9	59.0	49.5	(31)	
		MCDB	46.3	48.3	55.6	65.0	71.9	78.8	86.9	88.4	82.7	72.8	62.4	52.8	(32)	
	10%	WB	41.4	43.2	48.7	56.9	63.7	71.4	77.3	78.2	74.7	65.9	57.1	47.3	(33)	
		MCDB	44.8	46.7	52.7	62.3	70.1	76.8	85.4	87.0	81.1	70.9	60.6	50.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.6	13.2	15.2	14.9	12.3	11.9	12.9	12.0	12.8	12.3	10.8	(35)	
		MCDBR	12.3	14.1	19.1	20.3	20.0	15.8	14.9	14.7	13.9	14.3	13.6	13.2	(36)	
	5% WB	MCWBR	8.9	10.0	12.5	10.7	9.5	7.4	6.3	6.0	6.3	7.6	9.0	9.8	(37)	
		MCWBR	11.5	12.6	15.8	15.9	15.3	12.5	13.6	13.6	12.4	11.5	11.2	11.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.409	0.443	0.504	0.536	0.542	0.566	0.561	0.534	0.485	0.448	0.418	0.406	(40)	
		taud	2.090	2.018	1.896	1.871	1.919	1.931	1.999	2.066	2.171	2.188	2.172	2.119	(41)	
	Ebn at Noon		240	246	242	242	243	236	236	240	246	244	236	232	(42)	
		Edh at Noon	40	48	59	63	61	60	56	51	44	40	36	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		660	882	1204	1485	1638	1500	1531	1581	1252	1017	756	593	(44)	
	RadStd		46	77	93	137	175	134	223	154	112	95	65	51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page